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In [5]: import numpy as np
import matplotlib.pyplot as plt

alpha = 0.5
A = 5.0
omega = 2.0

t_max = 100
num_points = 5000
t = np.linspace(0, t_max, num_points)
dt = t[1] - t[0]

initial_conditions = [
    (1.0, 0.0),
    (-2.0, 1.0),
    (0.0, -3.0),
    (3.0, 3.0)
]

plt.figure(figsize=(10, 8))

for i, (x0, v0) in enumerate(initial_conditions):
    x = np.zeros_like(t)
    v = np.zeros_like(t)
    x[0], v[0] = x0, v0

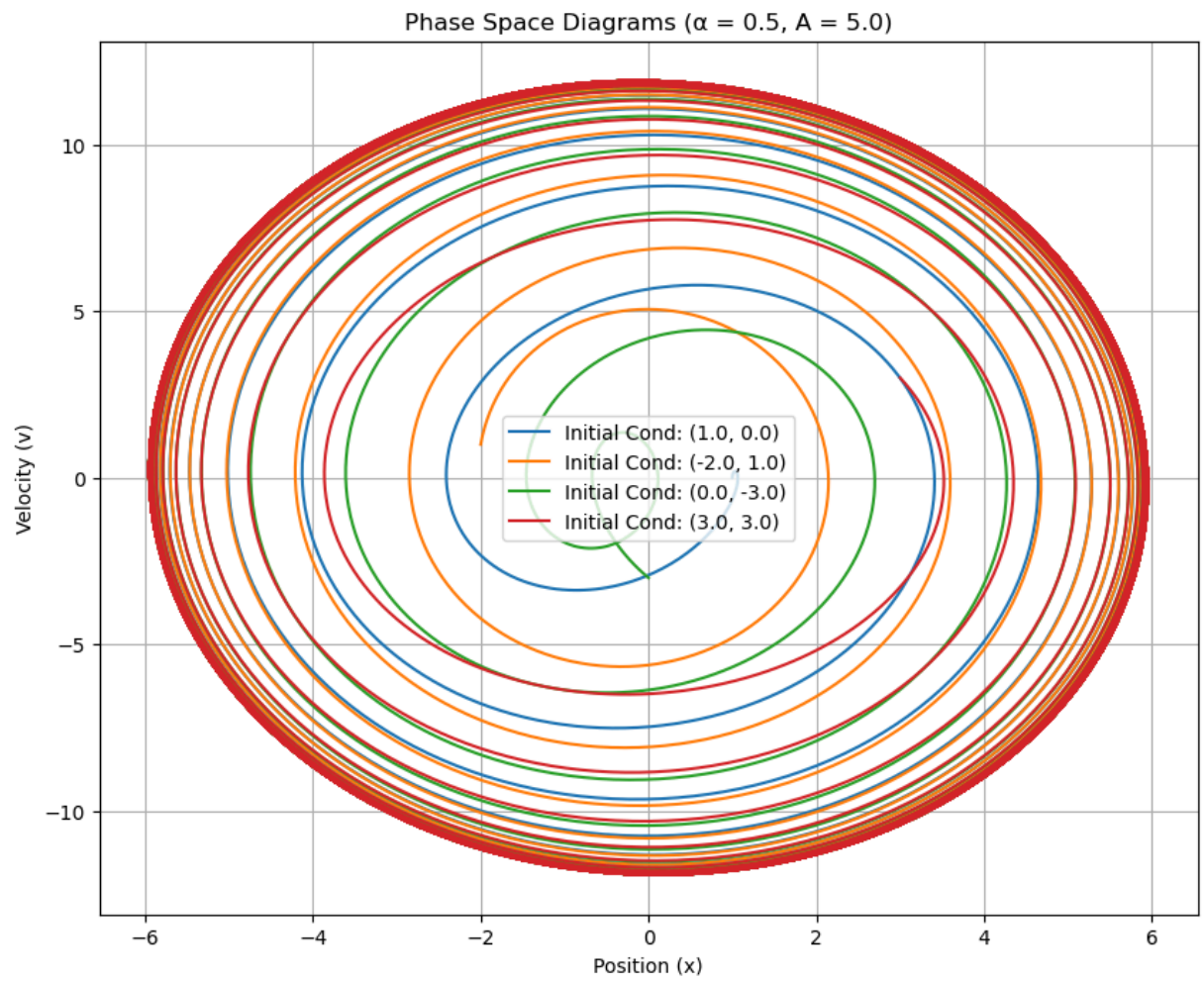
    for j in range(1, len(t)):
        forcing_term = A * np.cos(omega * t[j-1])

        x[j] = x[j-1] + v[j-1] * dt
        v[j] = v[j-1] + (-omega**2 * x[j-1] - alpha * v[j-1] + forcing_term) * dt

    plt.plot(x, v, label=f'Initial Cond: ({x0}, {v0})')

plt.xlabel('Position (x)')
plt.ylabel('Velocity (v)')
plt.title(f'Phase Space Diagrams ( $\alpha$  = {alpha}, A = {A})')
plt.legend()
plt.grid()

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In []: